

Myelomib 3.5mg Lyophilized Powder for Solution for Injection

1. NAME OF THE MEDICINAL PRODUCT

Myelomib 3.5mg Lyophilized Powder for Solution for Injection

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Each vial contains 3.5mg of Bortezomib.

For intravenous injection, each 8mL vial is reconstituted with 3.5mL of sodium chloride 9mg/mL (0.9%) solution, and the reconstituted solution is clear and colourless.

For subcutaneous injection, each 8mL vial is reconstituted with 1.4 mL of sodium chloride 9 mg/mL (0.9%) solution for injection, and the reconstituted solution is clear and colourless.

3. PHARMACEUTICAL FORM

Lyophilized powder for solution for injection.

White or off-white lyophilized powder or cake in colorless vial with grey rubber stopper and seal with aluminium flip-off cap.

The reconstituted solution should be clear and colourless.

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

Myelomib is indicated for the treatment of patients with multiple myeloma.

Myelomib is indicated for the treatment of patients with mantle cell lymphoma.

4.2 Posology and method of administration

Important Dosing Guidelines

Myelomib is for intravenous or subcutaneous use only. Do not administer Myelomib by any other route.

Because each route of administration has a different reconstituted concentration, use caution when calculating the volume to be administered.

The recommended starting dose of Myelomib is 1.3 mg/m². Myelomib may be administered intravenously at a concentration of 1 mg/mL, or subcutaneously at a concentration of 2.5 mg/mL.

Myelomib retreatment may be considered for patients with multiple myeloma who had previously responded to treatment with Myelomib and who have relapsed at least 6 months after completing prior Myelomib treatment. Treatment may be started at the last tolerated dose.

When administered intravenously, administer Myelomib as a 3 to 5 second bolus intravenous

injection.

Dosage in Previously Untreated Multiple Myeloma

Myelomib is administered in combination with oral melphalan and oral prednisone for 9, six-week treatment cycles as shown in Table 1. In Cycles 1 to 4, Myelomib is administered twice weekly (Days 1, 4, 8, 11, 22, 25, 29 and 32). In Cycles 5 to 9, Myelomib is administered once weekly (Days 1, 8, 22 and 29). At least 72 hours should elapse between consecutive doses of Myelomib.

Table 1: Dosage Regimen for Patients with Previously Untreated Multiple Myeloma

Twice Weekly Myelomib (Cycle 1 to 4)												
Week	1				2		3	4		5		6
Myelomib (1.3mg/m ²)	Day 1	-	-	Day 4	Day 8	Day 11	Rest period	Day 22	Day 25	Day 29	Day 32	Rest period
Melphalan (9mg/m ²)	Day 1	Day 2	Day 3	Day 4	-	-	Rest period	-	-	-	-	Rest period
Prednisone (60mg/m ²)	Day 1	Day 2	Day 3	Day 4	-	-	Rest period	-	-	-	-	Rest period
Once Weekly Myelomib (Cycles 5 to 9 when used in combination with Melphalan and Prednisone)												
Week	1				2		3	4		5		6
Myelomib (1.3mg/m ²)	Day 1	-	-		Day 8		Rest period	Day 22		Day 29		Rest period
Melphalan (9mg/m ²)	Day 1	Day 2	Day 3	Day 4	-	-	Rest period	-	-	-	-	Rest period
Prednisone (60mg/m ²)	Day 1	Day 2	Day 3	Day 4	-	-	Rest period	-	-	-	-	Rest period

Dose Modification Guidelines for Myelomib when given in Combination with Melphalan and Prednisone

Prior to initiating any cycle of therapy with Myelomib in combination with melphalan and prednisone:

- Platelet count should be at least $70 \times 10^9/L$ and the absolute neutrophil count (ANC) should be at least $1.0 \times 10^9/L$
- Nonhematological toxicities should have resolved to Grade 1 or baseline

Table 2: Dose Modifications during Cycles of Combination Myelomib, Melphalan and Prednisone Therapy

Toxicity	Dose Modification or Delay
Hematological toxicity during a cycle: If prolonged Grade 4 neutropenia or thrombocytopenia, or thrombocytopenia with bleeding is observed in the previous cycle	Consider reduction of the melphalan dose by 25% in the next cycle
If platelet count is not above $30 \times 10^9/L$ or ANC is not above $0.75 \times 10^9/L$ on a Myelomib dosing day (other than day 1)	Withhold Myelomib dose
If several Myelomib doses in consecutive cycles are withheld due to toxicity	Reduce Myelomib dose by 1 dose level (from $1.3\text{mg}/\text{m}^2$ to $1 \text{ mg}/\text{m}^2$, or from $1\text{mg}/\text{m}^2$ to $0.7\text{mg}/\text{m}^2$)

Toxicity	Dose Modification or Delay
Grade 3 or higher non-hematological toxicities	Withhold Myelomib therapy until symptoms of toxicity have resolved to Grade 1 or baseline. Then, Myelomib may be reinitiated with one dose level reduction (from 1.3mg/m ² to 1mg/m ² , or from 1mg/m ² to 0.7mg/m ²). For Myelomib-related neuropathic pain and/or peripheral neuropathy, hold or modify Myelomib as outlined in Table 6 .

For information concerning melphalan and prednisone, see manufacturer's prescribing information.

Dose modifications guidelines for peripheral neuropathy are provided.

Posology for previously untreated multiple myeloma patients eligible for haematopoietic stem cell transplantation (induction therapy)

Combination therapy with dexamethasone

Myelomib 3.5mg lyophilized powder for solution for injection is administered via intravenous or subcutaneous injection at the recommended dose of 1.3mg/m² body surface area twice weekly for two weeks on days 1, 4, 8, and 11 in a 21-day treatment cycle. This 3-week period is considered a treatment cycle. At least 72 hours should elapse between consecutive doses of Myelomib.

Dexamethasone is administered orally at 40 mg on days 1, 2, 3, 4, 8, 9, 10 and 11 of the Myelomib treatment cycle.

Four treatment cycles of this combination therapy are administered.

Combination therapy with dexamethasone and thalidomide

Myelomib 3.5mg lyophilized powder for solution for injection is administered via intravenous or subcutaneous injection at the recommended dose of 1.3 mg/m² body surface area twice weekly for two weeks on days 1, 4, 8, and 11 in a 28-day treatment cycle. This 4-week period is considered a treatment cycle. At least 72 hours should elapse between consecutive doses of Myelomib.

Dexamethasone is administered orally at 40 mg on days 1, 2, 3, 4, 8, 9, 10 and 11 of the Myelomib treatment cycle.

Thalidomide is administered orally at 50 mg daily on days 1-14 and if tolerated the dose is increased to 100 mg on days 15-28, and thereafter may be further increased to 200 mg daily from cycle 2.

Four treatment cycles of this combination are administered. It is recommended that patients with at least partial response receive 2 additional cycles.

Table 3: Posology for Myelomib combination therapy for patients with previously untreated multiple myeloma eligible for haematopoietic stem cell transplantation

Mb+Dx	Cycles 1 to 4			
	Week	1	2	3
	Mb (1.3mg/m ²)	Day 1, 4	Day 8, 11	Rest period
	Dx 40mg	Day 1, 2, 3, 4	Day 8, 9, 10, 11	-
Mb+Dx+T	Cycle 1			
	Week	1	2	3
				4

	Mb (1.3mg/m ²)	Day 1, 4	Day 8, 11	Rest period	Rest period
	T 50mg	Daily	Daily	-	-
	T 100mg ^a	-	-	Daily	Daily
	Dx 40mg	Day 1, 2, 3, 4	Day 8, 9, 10, 11	-	-
	Cycles 2 to 4^b				
	Mb (1.3mg/m ²)	Day 1, 4	Day 8, 11	Rest period	Rest period
	T 200mg ^a	Daily	Daily	Daily	Daily
	Dx 40mg	Day 1, 2, 3, 4	Day 8, 9, 10, 11	-	-

Mb=Myelomib, Dx=Dexamethasone, T=Thalidomide

- a Thalidomide dose is increased to 100 mg from week 3 of Cycle 1 only if 50 mg is tolerated and to 200 mg from cycle 2 onwards if 100 mg is tolerated.
- b Up to 6 cycles may be given to patients who achieve at least a partial response after 4 cycles

Dosage adjustments for transplant eligible patients

For Myelomib dosage adjustments, as described under “**Dosage and Dose Modifications for Relapsed Multiple Myeloma and Relapsed Mantle Cell Lymphoma**” and “**Dose Modifications for Peripheral Neuropathy**” should be followed.

In addition, when Myelomib is given in combination with other chemotherapeutic medicinal products, appropriate dose reductions for these products should be considered in the event of toxicities according to the recommendations in the local package inserts.

Dosage in Previously Untreated Mantle Cell Lymphoma

Myelomib (1.3mg/m²) is administered intravenously in combination with intravenous rituximab, cyclophosphamide, doxorubicin and oral prednisone (VcR-CAP) for 6 three week treatment cycles as shown in **Table 4**. Myelomib is administered first followed by rituximab. Myelomib is administered twice weekly for two weeks (Days 1, 4, 8, and 11) followed by a ten day rest period on Days 12 to 21. For patients with a response first documented at cycle 6, two additional VcR-CAP cycles are recommended. At least 72 hours should elapse between consecutive doses of Myelomib.

Table 4: Dosage Regimen for Patients with Previously Untreated Mantle Cell Lymphoma

Twice Weekly Myelomib (6, Three Week Cycles)*								
Week	1					2		3
Myelomib (1.3mg/m ²)	Day 1	-	-	Day 4	-	Day 8	Day 11	Rest period
Rituximab (375mg/m ²) Cyclophosphamide (750mg/m ²) Doxorubicin (50mg/m ²)	Day 1	-	-	-	-	-	-	Rest period
Prednisone (100mg/m ²)	Day 1	Day 2	Day 3	Day 4	Day 5	-	-	Rest period

* Dosing may continue for 2 more cycles (for a total of 8 cycles) if response is first seen

at cycle 6.

Dose Modification Guidelines for Myelomib When Given in Combination with Rituximab, Cyclophosphamide, Doxorubicin and Prednisone

Prior to the first day of each cycle (other than Cycle 1):

- Platelet count should be at least $100 \times 10^9/L$ and absolute neutrophil count (ANC) should be at least $1.5 \times 10^9/L$
- Hemoglobin should be at least 8g/dL (at least 4.96mmol/L)
- Non-hematologic toxicity should have recovered to Grade 1 or baseline

Interrupt Myelomib treatment at the onset of any Grade 3 hematologic or non-hematological toxicities, excluding neuropathy. For dose adjustments, see **Table 5** below.

Table 5: Dose Modifications on Days 4, 8, and 11 during Cycles of Combination Myelomib, Rituximab, Cyclophosphamide, Doxorubicin and Prednisone Therapy

Toxicity	Dose modification or delay
Hematological toxicity • Grade 3 or higher neutropenia, or a platelet count not at or above $25 \times 10^9/L$	Withhold Myelomib therapy for up to 2 weeks until the patient has an ANC at or above $0.75 \times 10^9/L$ and a platelet count at or above $25 \times 10^9/L$. • If, after Myelomib has been withheld, the toxicity does not resolve, discontinue Myelomib. • If toxicity resolves such that the patient has an ANC at or above $0.75 \times 10^9/L$ and a platelet count at or above $25 \times 10^9/L$, Myelomib dose should be reduced by 1 dose level (from $1.3\text{mg}/\text{m}^2$ to $1\text{mg}/\text{m}^2$, or from $1\text{mg}/\text{m}^2$ to $0.7\text{mg}/\text{m}^2$).
Grade 3 or higher non-hematological toxicities	Withhold Myelomib therapy until symptoms of the toxicity have resolved to Grade 2 or better. Then, Myelomib may be reinitiated with one dose level reduction (from $1.3\text{mg}/\text{m}^2$ to $1\text{mg}/\text{m}^2$, or from $1\text{mg}/\text{m}^2$ to $0.7\text{mg}/\text{m}^2$). For Myelomib-related neuropathic pain and/or peripheral neuropathy, hold or modify Myelomib as outlined in Table 6.

For information concerning rituximab, cyclophosphamide, doxorubicin and prednisone, see manufacturer's prescribing information.

Dosage and Dose Modifications for Relapsed Multiple Myeloma and Relapsed Mantle Cell Lymphoma

Myelomib ($1.3\text{mg}/\text{m}^2/\text{dose}$) is administered twice weekly for two weeks (Days 1, 4, 8, and 11) followed by a tenday rest period (Days 12 to 21). For extended therapy of more than eight cycles, Myelomib may be administered on the standard schedule or, for relapsed multiple myeloma, on a maintenance schedule of once weekly for four weeks (Days 1, 8, 15, and 22) followed by a 13 day rest period (Days 23 to 35). At least 72 hours should elapse between consecutive doses of Myelomib.

Patients with multiple myeloma who have previously responded to treatment with Myelomib (either alone or in combination) and who have relapsed at least six months after their prior Myelomib therapy may be started on Myelomib at the last tolerated dose. Retreated patients are administered Myelomib twice weekly (Days 1, 4, 8, and 11) every three weeks for a maximum of eight cycles. At least 72 hours should elapse between consecutive doses of Myelomib. Myelomib may be administered either as a single agent or in combination with dexamethasone.

Myelomib therapy should be withheld at the onset of any Grade 3 non-hematological or Grade 4 hematological toxicities excluding neuropathy. Once the symptoms of the toxicity have resolved, Myelomib therapy may be reinitiated at a 25% reduced dose (1.3 g/m²/dose reduced to 1mg/m²/dose; 1mg/m²/dose reduced to 0.7mg/m²/dose).

For dose modifications guidelines for peripheral neuropathy, see Dose modifications for peripheral neuropathy below.

Dose Modifications for Peripheral Neuropathy

Starting Myelomib subcutaneously may be considered for patients with pre-existing or at high risk of peripheral neuropathy. Patients with pre-existing severe neuropathy should be treated with Myelomib only after careful risk-benefit assessment.

Patients experiencing new or worsening peripheral neuropathy during Myelomib therapy may require a decrease in the dose and/or a less dose-intensive schedule.

For dose or schedule modification guidelines for patients who experience Myelomib-related neuropathic pain and/or peripheral neuropathy see Table 6.

Table 6: Recommended Dose Modification for Myelomib related Neuropathic Pain and/or Peripheral Sensory or Motor Neuropathy

Severity of Peripheral Neuropathy Signs and Symptoms*	Modification of Dose and Regimen
Grade 1 (asymptomatic; loss of deep tendon reflexes or paresthesias) without pain or loss of function	No action
Grade 1 with pain or Grade 2 (moderate symptoms; limiting instrumental Activities of Daily Living (ADL)**)	Reduce Myelomib to 1mg/m ²
Grade 2 with pain or Grade 3 (severe symptoms; limiting self-care ADL***)	Withhold Myelomib therapy until toxicity resolves. When toxicity resolves reinitiate with a reduced dose of Myelomib at 0.7mg/m ² once per week.
Grade 4 (life- threatening consequences; urgent intervention indicated)	Discontinue Myelomib

* Grading based on NCI Common Toxicity Criteria CTCAE v4.0.

** Instrumental ADL: refers to preparing meals, shopping for groceries or clothes, using telephone, managing money etc.

*** Self-care ADL: refers to bathing, dressing and undressing, feeding self, using the toilet, taking medications, and not bedridden

Special Populations

Patients with Renal Impairment

The pharmacokinetics of Myelomib are not influenced by the degree of renal impairment. Therefore, dosing adjustments of Myelomib are not necessary for patients with renal insufficiency. Since dialysis may reduce Myelomib concentrations, the drug should be administered after the dialysis procedure.

Patients with Hepatic Impairment

Patients with mild hepatic impairment do not require a starting dose adjustment and should be treated per the recommended Myelomib dose. For patients with moderate or severe hepatic impairment, see Table 7 below:

Table 7: Recommended Starting Dose Modification for Myelomib in Patients with Hepatic Impairment

	Bilirubin Level	SGOT (AST) Levels	Modification of Starting Dose in Multiple Myeloma and Relapsed Mantle Cell Lymphoma (1.3 mg/m ² twice weekly)
Mild	≤ 1.0x ULN	> ULN	None
	> 1.0x–1.5x ULN	Any	None
Moderate	> 1.5x–3x ULN	Any	Reduce Myelomib to 0.7mg/m ² in the first cycle. Consider dose escalation to 1.0mg/m ² or further dose reduction to 0.5mg/m ² in subsequent cycles based on patient tolerability.
Severe	> 3x ULN	Any	

Abbreviations: SGOT = serum glutamic oxaloacetic transaminase; AST = aspartate aminotransferase; ULN = upper limit of the normal range.

Administration

Myelomib is administered intravenously or subcutaneously. When administered intravenously, Myelomib is administered as a 3-5 second bolus intravenous injection through a peripheral or central intravenous catheter followed by a flush with 0.9% sodium chloride solution for injection. For subcutaneous administration, the reconstituted solution is injected into the thighs (right or left) or abdomen (right or left). Injection sites should be rotated for successive injections.

If local injection site reactions occur following Myelomib injection subcutaneously, a less concentrated Myelomib solution (1mg/ml instead of 2.5mg/ml) may be administered subcutaneously, or changed to IV injection.

4.3 Contraindications

Myelomib is contraindicated in patients with hypersensitivity to bortezomib or any of the excipients.

4.4 Special warnings and precautions for use

Myelomib should be administered under the supervision of a physician experienced in the use of antineoplastic therapy.

There have been fatal cases of inadvertent intrathecal administration of Bortezomib. Bortezomib is for IV and subcutaneous use only. **DO NOT ADMINISTER MYELOMIB INTRATHECALLY.**

Overall, the safety profile of patients treated with Bortezomib in monotherapy was similar to that observed in patients treated with Bortezomib in combination with melphalan and prednisone.

Peripheral Neuropathy

Bortezomib treatment causes a peripheral neuropathy (PN) that is predominantly sensory. However, cases of severe motor neuropathy with or without sensory peripheral neuropathy have been reported.

Patients with pre-existing symptoms (numbness, pain or a burning feeling in the feet or hands) and/or signs of peripheral neuropathy may experience worsening peripheral neuropathy (including $\geq$ Grade 3) during treatment with Bortezomib. Patients should be monitored for symptoms of neuropathy, such as a burning sensation, hyperesthesia, hypoesthesia, paresthesia, discomfort, neuropathic pain or weakness. In the study comparing Bortezomib IV vs SC the incidence of Grade ≥ 2 peripheral neuropathy events was 24% for SC and 41% for IV ($p=0.0124$). Grade ≥ 3 peripheral neuropathy occurred in 6% of subjects in the SC treatment group, compared with 16% in the IV treatment group ($p=0.0264$). Therefore, patients with pre-existing PN or at high risk of peripheral neuropathy may benefit from starting Bortezomib subcutaneously.

Patients experiencing new or worsening peripheral neuropathy may require a change in dose schedule or route of administration to SC. Following dose adjustments, improvement in or resolution of peripheral neuropathy was reported in 51% of patients with $\geq$ Grade 2 peripheral neuropathy in the single agent phase 3 multiple myeloma study of Bortezomib vs dexamethasone. Improvement in or resolution of peripheral neuropathy was reported in 73% of patients who discontinued due to Grade 2 neuropathy or who had $\geq$ Grade 3 peripheral neuropathy in the phase 2 multiple myeloma studies.

The long-term outcome of peripheral neuropathy has not been studied in mantle cell lymphoma.

Hypotension

In single agent multiple myeloma studies, the incidence of hypotension (postural, orthostatic, and Hypotension Not Otherwise Specified) was 11% to 12%. These events are observed throughout therapy. Caution should be used when treating patients with a history of syncope, patients receiving medications known to be associated with hypotension, and patients who are dehydrated. Management of orthostatic/postural hypotension may include adjustment of antihypertensive medications, hydration, or administration of mineralocorticoids and/or sympathomimetics.

Cardiac Disorders

Acute development or exacerbation of congestive heart failure, and/or new onset of decreased left ventricular ejection fraction has been reported, including reports in patients with few or no risk factors for decreased left ventricular ejection fraction. Patients with risk factors for, or existing heart disease should be closely monitored. In the single agent phase 3 multiple

myeloma study of Bortezomib vs dexamethasone, the incidence of any treatment-emergent cardiac disorder was 15% and 13% respectively. The incidence of heart failure events (acute pulmonary edema, cardiac failure, congestive cardiac failure, cardiogenic shock, pulmonary oedema) was similar in the Bortezomib and dexamethasone groups, 5% and 4%, respectively.

There have been isolated cases of QT-interval prolongation in clinical studies; causality has not been established.

Hepatic Events

Rare cases of acute liver failure have been reported in patients receiving multiple concomitant medications and with serious underlying medical conditions. Other reported hepatic events include increases in liver enzymes, hyperbilirubinemia, and hepatitis. Such changes may be reversible upon discontinuation of Bortezomib. There is limited re-challenge information in these patients.

Pulmonary Disorders

There have been rare reports of acute diffuse infiltrative pulmonary disease of unknown etiology such as pneumonitis, interstitial pneumonia, lung infiltration and Acute Respiratory Distress Syndrome (ARDS) in patients receiving Bortezomib. Some of these events have been fatal. A higher proportion of these events have been reported in Japan. In the event of new or worsening pulmonary symptoms, a prompt diagnostic evaluation should be performed and patients treated appropriately.

Two patients given high-dose cytarabine (2 g/m² per day) by continuous infusion with daunorubicin and Bortezomib for relapsed acute myelogenous leukemia died of ARDS early in the course of therapy.

Laboratory Tests

Complete blood counts (CBC) should be frequently monitored throughout treatment with Bortezomib.

Thrombocytopenia/Neutropenia

Bortezomib is associated with thrombocytopenia and neutropenia. Platelets were lowest at Day 11 of each cycle of Bortezomib treatment and typically recovered to baseline by the next cycle. The cyclical pattern of platelet count decrease and recovery remain consistent in the studies of multiple myeloma and mantle cell lymphoma, with no evidence of cumulative thrombocytopenia or neutropenia in any of the regimens studied.

Platelet counts should be monitored prior to each dose of Bortezomib. Bortezomib therapy should be held when the platelet count is <25,000/uL. There have been reports of gastrointestinal and intracerebral hemorrhage in association with Bortezomib. Transfusion and supportive care may be considered.

In the single-agent multiple myeloma study of Bortezomib vs dexamethasone, the mean platelet count nadir measured was approximately 40% of baseline. The severity of thrombocytopenia related to pretreatment platelet count is shown in Table 8. The incidence of significant bleeding events ($\geq$ Grade 3) was similar on both the Bortezomib (4%) and dexamethasone (5%) arms.

Table 8: Severity of Thrombocytopenia Related to Pretreatment Platelet Count in the

Single Agent Phase 3 Multiple Myeloma Study of Bortezomib vs Dexamethasone

Pretreatment Platelet Count ^a	Number of Patients (N = 331) ^b	Number (%) of Patients with Platelet Count < 10,000/ μ L	Number (%) of Patients with Platelet Count 10,000-25,000/ μ L
$\geq 75,000/\mu\text{L}$	309	8 (3%)	36 (12%)
$\geq 50,000/\mu\text{L} - < 75,000/\mu\text{L}$	14	2 (14%)	11 (79%)
$\geq 10,000/\mu\text{L} - < 50,000/\mu\text{L}$	7	1 (14%)	5 (71%)

^a A baseline platelet count of 50,000/ μ L was required for study eligibility.

^b Data were missing at baseline for 1 patient.

In the combination study of Bortezomib with rituximab, cyclophosphamide, doxorubicin and prednisone (VcR-CAP) in previously untreated mantle cell lymphoma patients, the incidence of thrombocytopenia adverse events ($\geq$ Grade 4) was 32% versus 2% for the rituximab, cyclophosphamide, doxorubicin, vincristine, and prednisone (R-CHOP) arm. The incidence of bleeding adverse events ($\geq$ Grade 3) was 1.7% (4 patients) in the VcR-CAP arm and was 1.2% (3 patients) in the R-CHOP arm.

There were no deaths due to bleeding events in either arm. There were no CNS bleeding events in the VcR-CAP arm; there was 1 bleeding event in the R-CHOP arm. Platelet transfusions were given to 23% of the patients in the VcR-CAP arm and 3% of the patients in the R-CHOP arm.

The incidence of neutropenia ($\geq$ Grade 4) was 70% in the VcR-CAP arm and was 52% in the R-CHOP arm. The incidence of febrile neutropenia ($\geq$ Grade 4) was 5% in the VcR-CAP arm and was 6% in the R-CHOP arm. Colony-stimulating factor support was provided at a rate of 78% in the VcR-CAP arm and 61% in the R-CHOP arm.

Gastrointestinal Adverse Events

Bortezomib treatment can cause nausea, diarrhea, constipation, and vomiting sometimes requiring use of antiemetics and antidiarrheal medications. Fluid and electrolyte replacement should be administered to prevent dehydration. Since patients receiving Bortezomib therapy may experience vomiting and/or diarrhea, patients should be advised regarding appropriate measures to avoid dehydration. Patients should be instructed to seek medical advice if they experience symptoms of dizziness, light headedness or fainting spells.

Tumor Lysis Syndrome

Because Bortezomib is a cytotoxic agent and can rapidly kill malignant cells the complications of tumor lysis syndrome may occur. The patients at risk of tumor lysis syndrome are those with high tumor burden prior to treatment. These patients should be monitored closely and appropriate precautions taken.

Patients with Hepatic Impairment

Bortezomib is metabolized by liver enzymes. Bortezomib exposure is increased in patients with moderate or severe hepatic impairment; these patients should be treated with Bortezomib at reduced starting doses and closely monitored for toxicities.

Posterior Reversible Encephalopathy Syndrome (PRES)

There have been reports of PRES in patients receiving Bortezomib. PRES is a rare, reversible, neurological disorder which can present with seizure, hypertension, headache,

lethargy, confusion, blindness, and other visual and neurological disturbances. Brain imaging, preferably MRI (Magnetic Resonance Imaging), is used to confirm the diagnosis. In patients developing PRES, discontinue Bortezomib. The safety of reinitiating Bortezomib therapy in patients previously experiencing PRES is not known.

4.5 Interaction with other medicinal products and other forms of interaction

Studies indicate that bortezomib is a weak inhibitor of cytochrome P450 (CYP) isozymes 1A2, 2C9, 2C19, 2D6 and 3A4. Based on the limited contribution (7%) of CYP2D6 to the metabolism of bortezomib, the CYP2D6 poor metabolizer phenotype is not expected to affect the overall disposition of bortezomib.

A drug-drug interaction study assessing the effect of ketoconazole, a potent CYP3A4 inhibitor, on the pharmacokinetics of Bortezomib, showed a bortezomib AUC mean increase of 35%, based on data from 12 patients. Therefore, patients should be closely monitored when given bortezomib in combination with potent CYP3A4-inhibitors (e.g., ketoconazole, ritonavir).

In a drug-drug interaction study assessing the effect of omeprazole, a potent inhibitor of CYP2C19, on the pharmacokinetics of Bortezomib there was no significant effect on the pharmacokinetics of bortezomib, based on data from 17 patients.

A drug-drug interaction study assessing the effect of rifampicin, a potent CYP3A4 inducer, on the pharmacokinetics of Bortezomib showed a mean bortezomib AUC reduction of 45% based on data from 6 patients. The concomitant use of Bortezomib with strong CYP3A4 inducers is therefore not recommended, as efficacy may be reduced. Examples of CYP3A4 inducers are rifampicin, carbamazepine, phenytoin, phenobarbital and St. John's Wort. In the same drug-drug interaction study, the effect of dexamethasone, a weaker CYP3A4 inducer was assessed. There was no significant effect on Bortezomib pharmacokinetics based on data from 7 patients.

A drug-drug interaction study assessing the effect of melphalan-prednisone on Bortezomib showed a 17% increase in mean bortezomib AUC based on data from 21 patients. This is not considered clinically relevant.

During clinical trials, hypoglycemia and hyperglycemia were reported in diabetic patients receiving oral hypoglycemics. Patients on oral antidiabetic agents receiving Bortezomib treatment may require close monitoring of their blood glucose levels and adjustment of the dose of their antidiabetic medication.

Patients should be cautioned about the use of concomitant medications that may be associated with peripheral neuropathy (such as amiodarone, anti-virals, isoniazid, nitrofurantoin, or statins), or with a decrease in blood pressure.

Drug Laboratory Test Interactions

None known.

4.6 Fertility, pregnancy and lactation

Contraception in males and females

Male and female patients of childbearing potential must use effective contraceptive measures during and for 3 months following treatment.

Pregnancy

No data are available for bortezomib with regard to exposure during pregnancy. The teratogenic potential of bortezomib has not been fully investigated.

Myelomib should not be used during pregnancy unless the clinical condition of the woman requires treatment with Myelomib.

If Myelomib is used during pregnancy, or if the patient becomes pregnant while receiving this medicinal product, the patient should be informed of potential for hazard to the foetus.

Thalidomide is a known human teratogenic active substance that causes severe life-threatening birth defects. Thalidomide is contraindicated during pregnancy and in women of childbearing potential unless all the conditions of the thalidomide pregnancy prevention programme are met. Patients receiving Myelomib in combination with thalidomide should adhere to the pregnancy prevention programme of thalidomide. Refer to the Summary of Product Characteristics of thalidomide for additional information.

Lactation

It is not known whether bortezomib is excreted in human milk. Because of the potential for serious adverse reactions in breast-fed infants, breast-feeding should be discontinued during treatment with Myelomib.

Fertility

Fertility studies were not conducted with Bortezomib.

4.7 Effects on ability to drive and use machines

Myelomib may have a moderate influence on the ability to drive and use machines. Myelomib may be associated with fatigue very commonly, dizziness commonly, syncope uncommonly and orthostatic/postural hypotension or blurred vision commonly. Therefore, patients must be cautious when driving or using machines and should be advised not to drive or operate machinery if they experience these symptoms.

4.8 Undesirable effects

Summary of the safety profile

Serious adverse reactions uncommonly reported during treatment with Bortezomib include cardiac failure, tumour lysis syndrome, pulmonary hypertension, posterior reversible encephalopathy syndrome, acute diffuse infiltrative pulmonary disorders and rarely autonomic neuropathy.

The most commonly reported adverse reactions during treatment with Bortezomib are nausea, diarrhoea, constipation, vomiting, fatigue, pyrexia, thrombocytopenia, anaemia, neutropenia, peripheral neuropathy (including sensory), headache, paraesthesia, decreased appetite, dyspnoea, rash, herpes zoster and myalgia.

Tabulated summary of adverse reactions

Multiple Myeloma

Table 9: Adverse reactions in patients with Multiple Myeloma treated with Bortezomib

System Organ Class	Incidence	Adverse reaction
Infections and infestations	Common	Herpes zoster (inc disseminated & ophthalmic), Pneumonia*, Herpes simplex*, Fungal infection*
	Uncommon	Infection*, Bacterial infections*, Viral infections*, Sepsis (inc septic shock)*, Bronchopneumonia, Herpes virus infection*, Meningoencephalitis herpetic [#] , Bacteraemia (inc staphylococcal), Hordeolum, Influenza, Cellulitis, Device related infection, Skin infection*, Ear infection*, Staphylococcal infection, Tooth infection*
	Rare	Meningitis (inc bacterial), Epstein-Barr virus infection, Genital herpes, Tonsillitis, Mastoiditis, Post viral fatigue syndrome
Neoplasms benign, malignant and unspecified (incl cysts and polyps)	Rare	Neoplasm malignant, Leukaemia plasmacytic, Renal cell carcinoma, Mass, Mycosis fungoides, Neoplasm benign*
Blood and lymphatic system disorders	Very Common	Thrombocytopenia*, Neutropenia*, Anaemia*
	Common	Leukopenia*, Lymphopenia*
	Uncommon	Pancytopenia*, Febrile neutropenia, Coagulopathy*, Leukocytosis*, Lymphadenopathy, Haemolytic anaemia [#]
	Rare	Disseminated intravascular coagulation, Thrombocytosis*, Hyperviscosity syndrome, Platelet disorder NOS, Thrombotic microangiopathy (inc thrombocytopenic purpura) [#] , Blood disorder NOS, Haemorrhagic diathesis, Lymphocytic infiltration
Immune system disorders	Uncommon	Angioedema [#] , Hypersensitivity*
	Rare	Anaphylactic shock, Amyloidosis, Type III immune complex mediated reaction
Endocrine disorders	Uncommon	Cushing's syndrome*, Hyperthyroidism*, Inappropriate antidiuretic hormone secretion
	Rare	Hypothyroidism
Metabolism and nutrition disorders	Very Common	Decreased appetite
	Common	Dehydration, Hypokalaemia*, Hyponatraemia*, Blood glucose abnormal*, Hypocalcaemia*, Enzyme abnormality*
	Uncommon	Tumour lysis syndrome, Failure to thrive*, Hypomagnesaemia*, Hypophosphataemia*, Hyperkalaemia*, Hypercalcaemia*, Hyponatraemia*, Uric acid abnormal*, Diabetes mellitus*, Fluid retention
	Rare	Hypermagnesaemia*, Acidosis, Electrolyte

System Organ Class	Incidence	Adverse reaction
		imbalance*, Fluid overload, Hypochloraemia*, Hypovolaemia, Hyperchloraemia*, Hyperphosphataemia*, Metabolic disorder, Vitamin B complex deficiency, Vitamin B12 deficiency, Gout, Increased appetite, Alcohol intolerance
Psychiatric disorders	Common	Mood disorders and disturbances*, Anxiety disorder*, Sleep disorders and disturbances*
	Uncommon	Mental disorder*, Hallucination*, Psychotic disorder*, Confusion*, Restlessness
	Rare	Suicidal ideation*, Adjustment disorder, Delirium, Libido decreased
Nervous system disorders	Very Common	Neuropathies*, Peripheral sensory neuropathy, Dysaesthesia*, Neuralgia*
	Common	Motor neuropathy*, Loss of consciousness (inc syncope), Dizziness*, Dysgeusia*, Lethargy, Headache*
	Uncommon	Tremor, Peripheral sensorimotor neuropathy, Dyskinesia*, Cerebellar coordination and balance disturbances*, Memory loss (exc dementia)*, Encephalopathy*, Posterior Reversible Encephalopathy Syndrome [#] , Neurotoxicity, Seizure disorders*, Post herpetic neuralgia, Speech disorder*, Restless legs syndrome, Migraine, Sciatica, Disturbance in attention, Reflexes abnormal*, Parosmia
	Rare	Cerebral haemorrhage*, Haemorrhage intracranial (inc subarachnoid)*, Brain oedema, Transient ischaemic attack, Coma, Autonomic nervous system imbalance, Autonomic neuropathy, Cranial palsy*, Paralysis*, Paresis*, Presyncope, Brain stem syndrome, Cerebrovascular disorder, Nerve root lesion, Psychomotor hyperactivity, Spinal cord compression, Cognitive disorder NOS, Motor dysfunction, Nervous system disorder NOS, Radiculitis, Drooling, Hypotonia, Guillain-Barré syndrome [#] , Demyelinating polyneuropathy [#]
Eye disorders	Common	Eye swelling*, Vision abnormal*, Conjunctivitis*
	Uncommon	Eye haemorrhage*, Eyelid infection*, Chalazion [#] , Blepharitis [#] , Eye inflammation*, Diplopia, Dry eye*, Eye irritation*, Eye pain, Lacrimation increased, Eye discharge
	Rare	Corneal lesion*, Exophthalmos, Retinitis, Scotoma, Eye disorder (inc. eyelid) NOS, Dacryoadenitis acquired, Photophobia, Photopsia, Optic neuropathy [#] , Different degrees of visual impairment (up to blindness)*
Ear and labyrinth disorders	Common	Vertigo*
	Uncommon	Dysacusis (inc tinnitus)*, Hearing impaired (up to and inc deafness), Ear discomfort*

System Organ Class	Incidence	Adverse reaction
	Rare	Ear haemorrhage, Vestibular neuronitis, Ear disorder NOS
Cardiac disorders	Uncommon	Cardiac tamponade [#] , Cardio-pulmonary arrest*, Cardiac fibrillation (inc atrial), Cardiac failure (inc left and right ventricular)*, Arrhythmia*, Tachycardia*, Palpitations, Angina pectoris, Pericarditis (inc pericardial effusion)*, Cardiomyopathy*, Ventricular dysfunction*, Bradycardia
	Rare	Atrial flutter, Myocardial infarction*, Atrioventricular block*, Cardiovascular disorder (inc cardiogenic shock), Torsade de pointes, Angina unstable, Cardiac valve disorders*, Coronary artery insufficiency, Sinus arrest
Vascular disorders	Common	Hypotension*, Orthostatic hypotension, Hypertension*
	Uncommon	Cerebrovascular accident [#] , Deep vein thrombosis*, Haemorrhage*, Thrombophlebitis (inc superficial), Circulatory collapse (inc hypovolaemic shock), Phlebitis, Flushing*, Haematoma (inc perirenal)*, Poor peripheral circulation*, Vasculitis, Hyperaemia (inc ocular)*
	Rare	Peripheral embolism, Lymphoedema, Pallor, Erythromelalgia, Vasodilatation, Vein discolouration, Venous insufficiency
Respiratory, thoracic and mediastinal disorders	Common	Dyspnoea*, Epistaxis, Upper/lower respiratory tract infection*, Cough*
	Uncommon	Pulmonary embolism, Pleural effusion, Pulmonary oedema (inc acute), Pulmonary alveolar haemorrhage [#] , Bronchospasm, Chronic obstructive pulmonary disease*, Hypoxaemia*, Respiratory tract congestion*, Hypoxia, Pleurisy*, Hiccups, Rhinorrhoea, Dysphonia, Wheezing
	Rare	Respiratory failure, Acute respiratory distress syndrome, Apnoea, Pneumothorax, Atelectasis, Pulmonary hypertension, Haemoptysis, Hyperventilation, Orthopnoea, Pneumonitis, Respiratory alkalosis, Tachypnoea, Pulmonary fibrosis, Bronchial disorder*, Hypocapnia*, Interstitial lung disease, Lung infiltration, Throat tightness, Dry throat, Increased upper airway secretion, Throat irritation, Upper-airway cough syndrome
Gastrointestinal disorders	Very Common	Nausea and vomiting symptoms*, Diarrhoea*, Constipation
	Common	Gastrointestinal haemorrhage (inc mucosal)*, Dyspepsia, Stomatitis*, Abdominal distension, Oropharyngeal pain*, Abdominal pain (inc gastrointestinal and splenic pain)*, Oral disorder*, Flatulence
	Uncommon	Pancreatitis (inc chronic)*, Haematemesis, Lip swelling*, Gastrointestinal obstruction (inc small

System Organ Class	Incidence	Adverse reaction
		intestinal obstruction, ileus)*, Abdominal discomfort, Oral ulceration*, Enteritis*, Gastritis*, Gingival bleeding, Gastroesophageal reflux disease*, Colitis (inc clostridium difficile)*, Colitis ischaemic [#] , Gastrointestinal inflammation*, Dysphagia, Irritable bowel syndrome, Gastrointestinal disorder NOS, Tongue coated, Gastrointestinal motility disorder*, Salivary gland disorder*
	Rare	Pancreatitis acute, Peritonitis*, Tongue oedema*, Ascites, Oesophagitis, Cheilitis, Faecal incontinence, Anal sphincter atony, Faecaloma*, Gastrointestinal ulceration and perforation*, Gingival hypertrophy, Megacolon, Rectal discharge, Oropharyngeal blistering*, Lip pain, Periodontitis, Anal fissure, Change of bowel habit, Proctalgia, Abnormal faeces
Hepatobiliary disorders	Common	Hepatic enzyme abnormality*
	Uncommon	Hepatotoxicity (inc liver disorder), Hepatitis*, Cholestasis
	Rare	Hepatic failure, Hepatomegaly, Budd-Chiari syndrome, Cytomegalovirus hepatitis, Hepatic haemorrhage, Cholelithiasis
Skin and subcutaneous tissue disorders	Common	Rash*, Pruritus*, Erythema, Dry skin
	Uncommon	Erythema multiforme, Urticaria, Acute febrile neutrophilic dermatosis, Toxic skin eruption, Toxic epidermal necrolysis [#] , Stevens-Johnson syndrome [#] , Dermatitis*, Hair disorder*, Petechiae, Ecchymosis, Skin lesion, Purpura, Skin mass*, Psoriasis, Hyperhidrosis, Night sweats, Decubitus ulcer [#] , Acne*, Blister*, Pigmentation disorder*
	Rare	Skin reaction, Jessner's lymphocytic infiltration, Palmar-plantar erythrodysesthesia syndrome, Haemorrhage subcutaneous, Livedo reticularis, Skin induration, Papule, Photosensitivity reaction, Seborrhoea, Cold sweat, Skin disorder NOS, Erythrosis, Skin ulcer, Nail disorder
Musculoskeletal and connective tissue disorders	Very Common	Musculoskeletal pain*
	Common	Muscle spasms*, Pain in extremity, Muscular weakness
	Uncommon	Muscle twitching, Joint swelling, Arthritis*, Joint stiffness, Myopathies*, Sensation of heaviness
	Rare	Rhabdomyolysis, Temporomandibular joint syndrome, Fistula, Joint effusion, Pain in jaw, Bone disorder, Musculoskeletal and connective tissue infections and inflammations*, Synovial cyst
Renal and urinary disorders	Common	Renal impairment*
	Uncommon	Renal failure acute, Renal failure chronic*, Urinary tract infection*, Urinary tract signs and symptoms*, Haematuria*, Urinary retention, Micturition disorder*,

System Organ Class	Incidence	Adverse reaction
		Proteinuria, Azotaemia, Oliguria*, Pollakiuria
	Rare	Bladder irritation
Reproductive system and breast disorders	Uncommon	Vaginal haemorrhage, Genital pain*, Erectile dysfunction,
	Rare	Testicular disorder*, Prostatitis, Breast disorder female, Epididymal tenderness, Epididymitis, Pelvic pain, Vulval ulceration
Congenital, familial and genetic disorders	Rare	Aplasia, Gastrointestinal malformation, Ichthyosis
General disorders and administration site conditions	Very Common	Pyrexia*, Fatigue, Asthenia
	Common	Oedema (inc peripheral), Chills, Pain*, Malaise*
	Uncommon	General physical health deterioration*, Face oedema*, Injection site reaction*, Mucosal disorder*, Chest pain, Gait disturbance, Feeling cold, Extravasation*, Catheter related complication*, Change in thirst*, Chest discomfort, Feeling of body temperature change*, Injection site pain*
	Rare	Death (inc sudden), Multi-organ failure, Injection site haemorrhage*, Hernia (inc hiatus)*, Impaired healing*, Inflammation, Injection site phlebitis*, Tenderness, Ulcer, Irritability, Non-cardiac chest pain, Catheter site pain, Sensation of foreign body
Investigations	Common	Weight decreased
	Uncommon	Hyperbilirubinaemia*, Protein analyses abnormal*, Weight increased, Blood test abnormal*, C-reactive protein increased
	Rare	Blood gases abnormal*, Electrocardiogram abnormalities (inc QT prolongation)*, International normalised ratio abnormal*, Gastric pH decreased, Platelet aggregation increased, Troponin I increased, Virus identification and serology*, Urine analysis abnormal*
Injury, poisoning and procedural complications	Uncommon	Fall, Contusion
	Rare	Transfusion reaction, Fractures*, Rigors*, Face injury, Joint injury*, Burns, Laceration, Procedural pain, Radiation injuries*
Surgical and medical procedures	Rare	Macrophage activation

NOS=not otherwise specified

* Grouping of more than one MedDRA preferred term.

Post-marketing adverse reaction regardless of indication

Mantle Cell Lymphoma (MCL)

Table 10: Adverse reactions in patients with Mantle Cell Lymphoma treated with VcR-CAP

System Organ Class	Incidence	Adverse reaction
Infections and infestations	Very Common	Pneumonia*
	Common	Sepsis (inc septic shock)*, Herpes zoster (inc disseminated & ophthalmic), Herpes virus infection*, Bacterial infections*, Upper/lower respiratory tract infection*, Fungal infection*, Herpes simplex*
	Uncommon	Hepatitis B, Infection*, Bronchopneumonia
Blood and lymphatic system disorders	Very Common	Thrombocytopenia*, Febrile neutropenia, Neutropenia*, Leukopenia*, Anaemia*, Lymphopenia*
	Uncommon	Pancytopenia*
Immune system disorders	Common	Hypersensitivity*
	Uncommon	Anaphylactic reaction
Metabolism and nutrition disorders	Very Common	Decreased appetite
	Common	Hypokalaemia*, Blood glucose abnormal*, Hyponatraemia*, Diabetes mellitus*, Fluid retention
	Uncommon	Tumour lysis syndrome
Psychiatric disorders	Common	Sleep disorders and disturbances*
Nervous system disorders	Very Common	Peripheral sensory neuropathy, Dysaesthesia*, Neuralgia*
	Common	Neuropathies*, Motor neuropathy*, Loss of consciousness (inc syncope), Encephalopathy*, Peripheral sensorimotor neuropathy, Dizziness*, Dysgeusia*, Autonomic neuropathy
	Uncommon	Autonomic nervous system imbalance
Eye disorders	Common	Vision abnormal*
Ear and labyrinth disorders	Common	Dysacusis (inc tinnitus)*
	Uncommon	Vertigo*, Hearing impaired (up to and inc deafness)
Cardiac disorders	Common	Cardiac fibrillation (inc atrial), Arrhythmia*, Cardiac failure (inc left and right ventricular)*, Myocardial ischaemia, Ventricular dysfunction*
	Uncommon	Cardiovascular disorder (inc cardiogenic shock)
Vascular disorders	Common	Hypertension*, Hypotension*, Orthostatic hypotension
Respiratory, thoracic and mediastinal disorders	Common	Dyspnoea*, Cough*, Hiccups
	Uncommon	Acute respiratory distress syndrome, Pulmonary embolism, Pneumonitis, Pulmonary hypertension, Pulmonary oedema (inc acute)
Gastrointestinal disorders	Very Common	Nausea and vomiting symptoms*, Diarrhoea*, Stomatitis*, Constipation
	Common	Gastrointestinal haemorrhage (inc mucosal)*, Abdominal distension, Dyspepsia, Oropharyngeal pain*, Gastritis*, Oral ulceration*, Abdominal discomfort, Dysphagia, Gastrointestinal inflammation*, Abdominal pain (inc gastrointestinal and splenic pain)*, Oral disorder*
	Uncommon	Colitis (inc clostridium difficile)*

System Organ Class	Incidence	Adverse reaction
Hepatobiliary disorders	Common	Hepatotoxicity (inc liver disorder)
	Uncommon	Hepatic failure
Skin and subcutaneous tissue disorders	Very Common	Hair disorder*
	Common	Pruritus*, Dermatitis*, Rash*
Musculoskeletal and connective tissue disorders	Common	Muscle spasms*, Musculoskeletal pain*, Pain in extremity
Renal and urinary disorders	Common	Urinary tract infection*
General disorders and administration site conditions	Very Common	Pyrexia*, Fatigue, Asthenia
	Common	Oedema (inc peripheral), Chills, Injection site reaction*, Malaise*
Investigations	Common	Hyperbilirubinaemia*, Protein analyses abnormal*, Weight decreased, Weight increased

* Grouping of more than one MedDRA preferred term.

Description of selected adverse reactions

Herpes zoster virus reactivation

Multiple Myeloma

The incidence of herpes zoster among patients in the Bortezomib+Melphalan+Prednisone treatment group was 17% for patients not administered antiviral prophylaxis compared to 3% for patients administered antiviral prophylaxis.

Mantle cell lymphoma

The incidence of herpes zoster among patients in the VcR-CAP arm was 10.7% for patients not administered antiviral prophylaxis compared to 3.6% for patients administered antiviral prophylaxis.

Hepatitis B Virus (HBV) reactivation and infection

Mantle cell lymphoma

The overall incidence of hepatitis B infections was similar in patients treated with VcR-CAP or with R-CHOP.

Peripheral neuropathy in combination regimens

Multiple Myeloma

The incidence of peripheral neuropathy in the combination regimens is presented in the table below:

Table 11: Incidence of peripheral neuropathy during induction treatment by toxicity

and treatment discontinuation due to peripheral neuropathy

	VDDx	VcDx	TDx	VcTDx
Incidence of PN (%)				
All Grade PN	3	15	12	45
≥ Grade 2 PN	1	10	2	31
≥ Grade 3 PN	< 1	5	0	5
Discontinuation due to PN (%)	< 1	2	1	5

VDDx=vincristine, doxorubicin, dexamethasone; VcDx=bortezomib, dexamethasone; TDx=thalidomide, dexamethasone; VcTDx=bortezomib, thalidomide, dexamethasone; PN=peripheral neuropathy

Note: Peripheral neuropathy included the preferred terms: neuropathy peripheral, peripheral motor neuropathy, peripheral sensory neuropathy, and polyneuropathy.

Mantle cell lymphoma

The incidence of peripheral neuropathy in the combination regimens is presented in the table below:

Table 12: Incidence of peripheral neuropathy by toxicity and treatment discontinuation due to peripheral neuropathy

	VcR-CAP	R-CHOP
Incidence of PN (%)		
All GradePN	30	29
≥ Grade 2 PN	18	9
≥ Grade 3 PN	8	4
Discontinuation due to PN (%)	2	< 1

VcR-CAP=bortezomib, rituximab, cyclophosphamide, doxorubicin, and prednisone; R-CHOP= rituximab, cyclophosphamide, doxorubicin, vincristine, and prednisone; PN=peripheral neuropathy

Peripheral neuropathy included the preferred terms: peripheral sensory neuropathy, neuropathy peripheral, peripheral motor neuropathy, and peripheral sensorimotor neuropathy

Elderly MCL patients

Although in patients aged ≥ 75 years, both VcR-CAP and R-CHOP were less tolerated, the serious adverse event rate in the VcR-CAP groups was 68%, compared to 42% in the R-CHOP group.

Notable differences in the safety profile of Bortezomib administered subcutaneously versus intravenously as single agent

Patients who received Bortezomib subcutaneously compared to intravenous administration had 13% lower overall incidence of treatment emergent adverse reactions that were Grade 3 or higher in toxicity, and a 5% lower incidence of discontinuation of Bortezomib. The overall incidence of diarrhoea, gastrointestinal and abdominal pain, asthenic conditions, upper

respiratory tract infections and peripheral neuropathies were 12%-15% lower in the subcutaneous group than in the intravenous group. In addition, the incidence of Grade 3 or higher peripheral neuropathies was 10% lower, and the discontinuation rate due to peripheral neuropathies 8% lower for the subcutaneous group as compared to the intravenous group.

Retreatment of patients with relapsed multiple myeloma

The most common all-grade adverse events occurring in patients were thrombocytopenia, neuropathy, anaemia, diarrhoea, and constipation. All grade peripheral neuropathy and grade ≥ 3 peripheral neuropathy were observed in patients.

4.9 Overdose

Overdosage more than twice the recommended dose in patients has been associated with the acute onset of symptomatic hypotension and thrombocytopenia with fatal outcomes.

There is no known specific antidote for Myelomib overdose. In the event of an overdose, the patient's vital signs should be monitored and appropriate supportive care given to maintain blood pressure (such as fluids, pressors, and/or inotropic agents) and body temperature.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Mechanism of Action

Bortezomib is a reversible inhibitor of the chymotrypsin-like activity of the 26S proteasome in mammalian cells. The 26S proteasome is a large protein complex that degrades ubiquitinated proteins. The ubiquitin-proteasome pathway plays an essential role in regulating the intracellular concentration of specific proteins, thereby maintaining homeostasis within cells. Inhibition of the 26S proteasome prevents this targeted proteolysis which can affect multiple signaling cascades within the cell. This disruption of normal homeostatic mechanisms can lead to cell death. Experiments have demonstrated that bortezomib is cytotoxic to a variety of cancer cell types *in vitro*. Bortezomib causes a delay in tumor growth *in vivo* in nonclinical tumor models, including multiple myeloma.

5.2 Pharmacokinetic properties

Pharmacokinetics

Following intravenous bolus administration of a 1.0 mg/m² and 1.3 mg/m² dose to eleven patients with multiple myeloma, the mean first-dose maximum plasma concentrations of bortezomib were 57 and 112 ng/mL, respectively. In subsequent doses, mean maximum observed plasma concentrations ranged from 67 to 106 ng/mL for the 1.0 mg/m² dose and 89 to 120 ng/mL for the 1.3 mg/m² dose. The mean elimination half-life of bortezomib upon multiple dosing ranged from 40 – 193 hours. Bortezomib is eliminated more rapidly following the first dose compared to subsequent doses. Mean total body clearances were 102 and 112 L/h following the first dose for doses of 1.0 mg/m² and 1.3 mg/m², respectively, and ranged from 15 to 32 L/h following subsequent doses for doses of 1.0 mg/m² and 1.3 mg/m², respectively.

In the PK/PD substudy in Phase 3 trial, following an IV bolus or subcutaneous (SC) injection of a 1.3 mg/m² dose to multiple myeloma patients (n = 14 for IV, n = 17 for SC), the total systemic exposure after repeat dose administration (AUC_{last}) was equivalent for SC and IV administration. The C_{max} after SC administration (20.4 ng/mL) was lower than IV (223 ng/mL). The AUC_{last} geometric mean ratio was 0.99 and 90% confidence intervals were 80.18% - 122.80%.

Distribution

The mean distribution volume of bortezomib ranged from 1659 liters to 3294 liters single- or repeat dose IV administration of 1.0mg/m² or 1.3mg/m² to patients with multiple myeloma. This suggests that bortezomib distributes widely to peripheral tissues. The binding of bortezomib to human plasma proteins averaged 83% over the concentration range of 100 - 1000ng/mL.

Metabolism

In vitro studies with human liver microsomes and human cDNA-expressed cytochrome P450 isozymes indicate that bortezomib is primarily oxidatively metabolized via cytochrome P450 enzymes, 3A4, 2C19, and 1A2. Bortezomib metabolism by CYP 2D6 and 2C9 enzymes is minor. The major metabolic pathway is deboronation to form two deboronated metabolites that subsequently undergo hydroxylation to several metabolites. Deboronated-bortezomib metabolites are inactive as 26S proteasome inhibitors. Pooled plasma data from 8 patients at 10min and 30min after dosing indicate that the plasma levels of metabolites are low compared to the parent drug.

Elimination

The pathways of elimination of Bortezomib have not been characterized in humans.

Special Populations

Age, Gender, and Race

The pharmacokinetics of Bortezomib were characterized following twice weekly intravenous bolus administration of 1.3 mg/m² doses to 104 pediatric patients (2-16 years old) with acute lymphoblastic leukemia (ALL) or acute myeloid leukemia (AML). Based on a population pharmacokinetic analysis, clearance of bortezomib increased with increasing body surface area (BSA). Geometric mean (%CV) clearance was 7.79 (25%) L/hr/m², volume of distribution at steady-state was 834 (39%) L/m², and the elimination half-life was 100 (44%) hours. After correcting for the BSA effect, other demographics such as age, body weight and sex did not have clinically significant effects on bortezomib clearance. BSA-normalized clearance of bortezomib in pediatric patients was similar to that observed in adults.

The effects of gender, and race on the pharmacokinetics of bortezomib have not been evaluated.

Hepatic Impairment

The effect of hepatic impairment on the pharmacokinetics of IV bortezomib was assessed in 60 cancer patients at bortezomib doses ranging from 0.5 to 1.3 mg/m². When compared to patients with normal hepatic function, mild hepatic impairment did not alter dose-normalised bortezomib AUC. However, the dose normalized mean AUC values were increased by approximately 60% in patients with moderate or severe hepatic impairment. A lower starting dose is recommended in patients with moderate or severe hepatic impairment, and those patients should be monitored closely.

Renal Impairment

A pharmacokinetic study was conducted in patients with various degrees of renal impairment who were classified according to their creatinine clearance values (CrCL) into the following groups: Normal (CrCL $\geq$ 60 mL/min/1.73 m², n = 12), Mild (CrCL = 40 - 59 mL/min/1.73 m², n = 10), Moderate (CrCL = 20 - 39 mL/min/1.73 m², n = 9), and Severe (CrCL < 20 mL/min/1.73 m², n = 3). A group of dialysis patients who were dosed after dialysis was also included in the study (n = 8). Patients were administered intravenous doses of 0.7 to 1.3 mg/m² of bortezomib twice weekly. Exposure of bortezomib (dose-normalized AUC and C_{max}) was comparable among all the groups.

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Mannitol, Tert-butyl alcohol, Water for injection, Nitrogen

6.2 Incompatibilities

This product must not be mixed with other medicinal products.

6.3 Shelf life

Please refer to the expiry date on the product labels.

Reconstituted solution: Use immediately after preparation. If the reconstituted solution is not used immediately, in-use storage times and conditions prior to use are the responsibility of the user. However, the reconstituted solution is stable for 8 hours at 25°C stored in the original vial and/or a syringe, with a total storage time for the reconstituted medicine not exceeding 8 hours prior to administration.

6.4 Special precautions for storage

Store below 30°C. Keep the vial in the outer carton in order to protect from light.

Reconstituted solution: Use immediately after preparation. If the reconstituted solution is not used immediately, in-use storage times and conditions prior to use are the responsibility of the user. However, the reconstituted solution is stable for 8 hours at 25°C stored in the original vial and/or a syringe, with a total storage time for the reconstituted medicine not exceeding 8 hours prior to administration.

6.5 Nature and contents of container

8ml Type I clear glass vial and sealed with a grey 20-D₂ lyophilization bromobutyl rubber stopper for injection, and secured with an aluminum-plastic flip-off cap, packed with outer carton with package insert.

Available in packs of 1 vial per carton.

6.6 Instruction for Use and Handling and Disposal

Administration Precautions

Bortezomib is an antineoplastic. Caution should be used during handling and preparation. Proper aseptic technique should be used. Use of gloves and other protective clothing to prevent skin contact is recommended. In clinical trials, local skin irritation was reported in 5% of patients, but extravasation of Bortezomib was not associated with tissue damage.

When administered subcutaneously, alternate sites for each injection (thigh or abdomen). New injections should be given at least one inch from an old site and never into areas where the site is tender, bruised, red, or hard.

There have been fatal cases of inadvertent intrathecal administration of Bortezomib. Bortezomib is for IV and subcutaneous use only. **DO NOT ADMINISTER MYELOMIB INTRATHECALLY.**

Reconstitution/Preparation for Intravenous and Subcutaneous Administration

The contents of each vial should be reconstituted only with 0.9% sodium chloride according to the following instructions based on route of administration:

	IV	SC
	3.5mg Bortezomib	3.5mg Bortezomib
Volume of diluent (0.9% Sodium Chloride) added to reconstitute one vial	3.5ml	1.4ml
Final Concentration after reconstitution (mg/ml)	1.0mg/ml	2.5mg/ml

The reconstituted product should be a clear and colorless solution. Parenteral drug products should be inspected visually for particulate matter and discoloration prior to administration whenever solution and container permit. If any discoloration or particulate matter is observed, the reconstituted product should not be used.

Procedure for Proper Disposal

Any unused product or waste material should be disposed of in accordance with local requirements.

7 MANUFACTURER

Manufactured by:

Jiangsu Hansoh Pharmaceutical Group Co., Ltd.,
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Lianyungang, Jiangsu 222069, China

Product Registration Holder:

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8 DATE OF REVISION

06/07/2023

Package insert is drafted in Times New Roman with font size of 12, subject to change based on any other legible font and font size.
